MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, LOW-POWER SCHOTTKY TTL,
4-BIT BINARY FULL ADDERS WITH FAST CARRY,
MONOLITHIC SILICON

This amendment forms a part of Military Specification MIL-M-38510/312B, dated 18 August 1983, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.2.3, Letter: Delete "X" and substitute "2".

1.3, Thermal resistance, junction to case (θ_{JC}), delete and substitute:

"Thermal resistance, junction-to-case (θ_{JC}):

Cases E and F - - - - - - - - - - (See MIL-M-38510, appendix C)
Case 2 - - - - - - - - - - - - - - 60°C/W 2/"

After Junction temperature (T_J), add: "3/".

After footnote 1/, add the following:

"2/ When a thermal resistance value is included in MIL-M-38510, appendix C, it shall supersede the value stated herein.

"3/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions per method 5004 of MIL-STD-883."

PAGE 2

3.2.4, delete in its entirety and substitute:

"3.2.4 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request."

3.6, delete in its entirety and substitute:

"3.6 Marking. Marking shall be in accordance with MIL-M-38510."

PAGE 4

TABLE I, t_{PHL5}, conditions column: Delete "R_L = 560Ω" and substitute "R_L = 2 kΩ".

TABLE I, t_{PLH5}, conditions column: Delete "R_L = 560Ω" and substitute "R_L = 2 kΩ".

PAGE 5

TABLE II, Group A test requirements, Class B devices, add: "8".

PAGE 6

4.2c, delete and substitute: "c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510."

4.4.3b(3), delete: "appendix B of MIL-M-38510 and".

FSC 5962

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PAGE 7

FIGURE 1: Delete "Case X" and substitute "Case 2".

PAGE 8

FIGURE 1: Delete "Case X" and substitute "Case 2".

PAGE 11

FIGURE 3: Delete and substitute new figure 3 as printed on page 3 of this amendment.

PAGE 12

FIGURE 4, Note 1, delete and substitute:

"1. CP = 100 kHz ±50%; duty cycle = 50 ±15%; V_{IH} = 2.0 V minimum to 5.5 V maximum; V_{IL} = -0.5 V minimum to 0.7 V maximum."

PAGE 13

FIGURE 5: Delete and substitute new figure 5 as printed on page 4 of this amendment.

PAGE 14

TABLE III, table heading: Delete "Case X" and substitute "Case 2". Make this same change on pages 15 through 19.

PAGE 16

TABLE III, footnote 1/: Delete "Case X" and substitute "Case 2".

TABLE III, footnote 5/, delete and substitute: "5/ A ≥ 2.5 V; B ≤ 0.4 V."

PAGE 19

TABLE III, footnote 5/, delete and substitute: "5/ A ≥ 2.0 V; B ≤ 0.4 V."

PAGE 20

6.1: Delete in its entirety.

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Device types 01 and 02

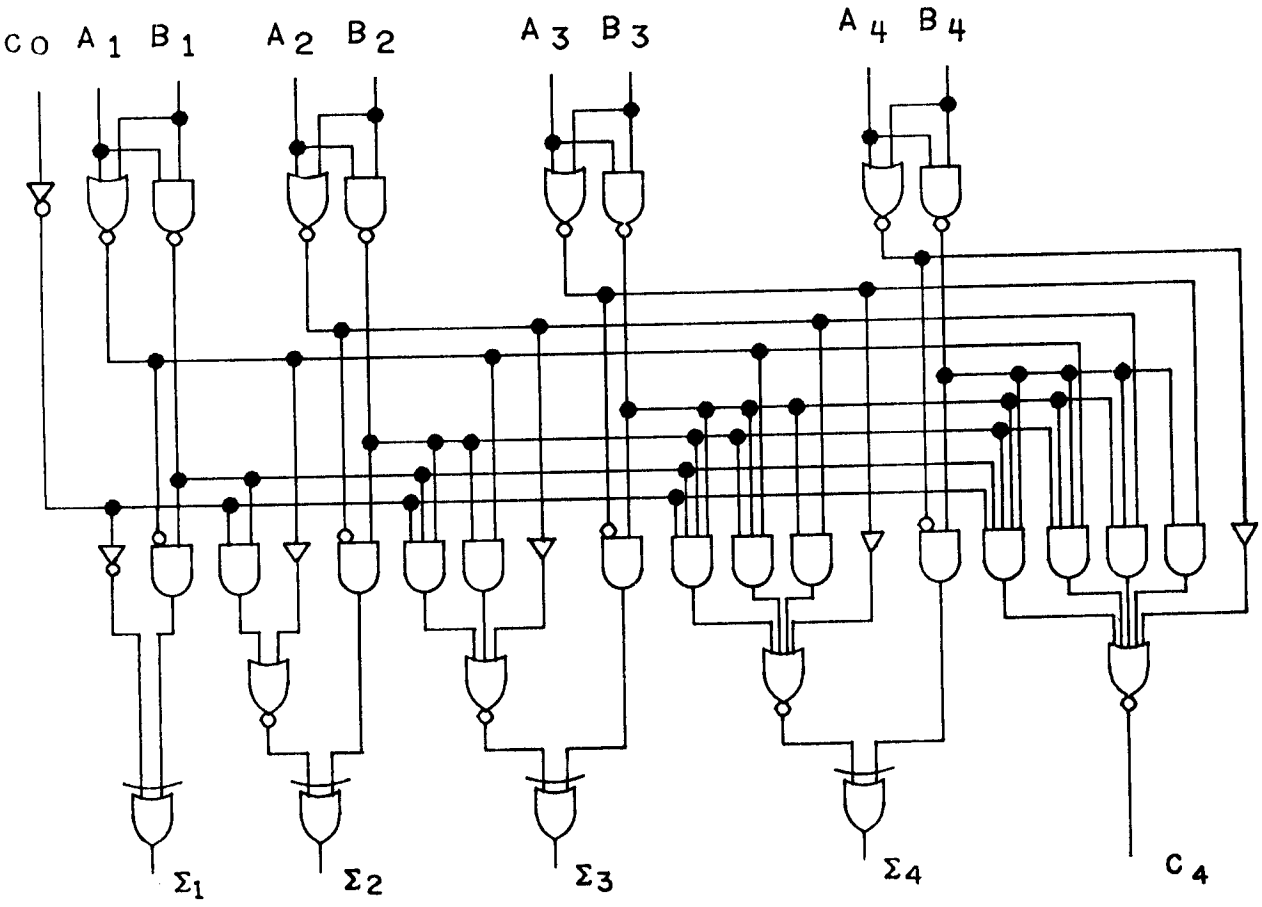
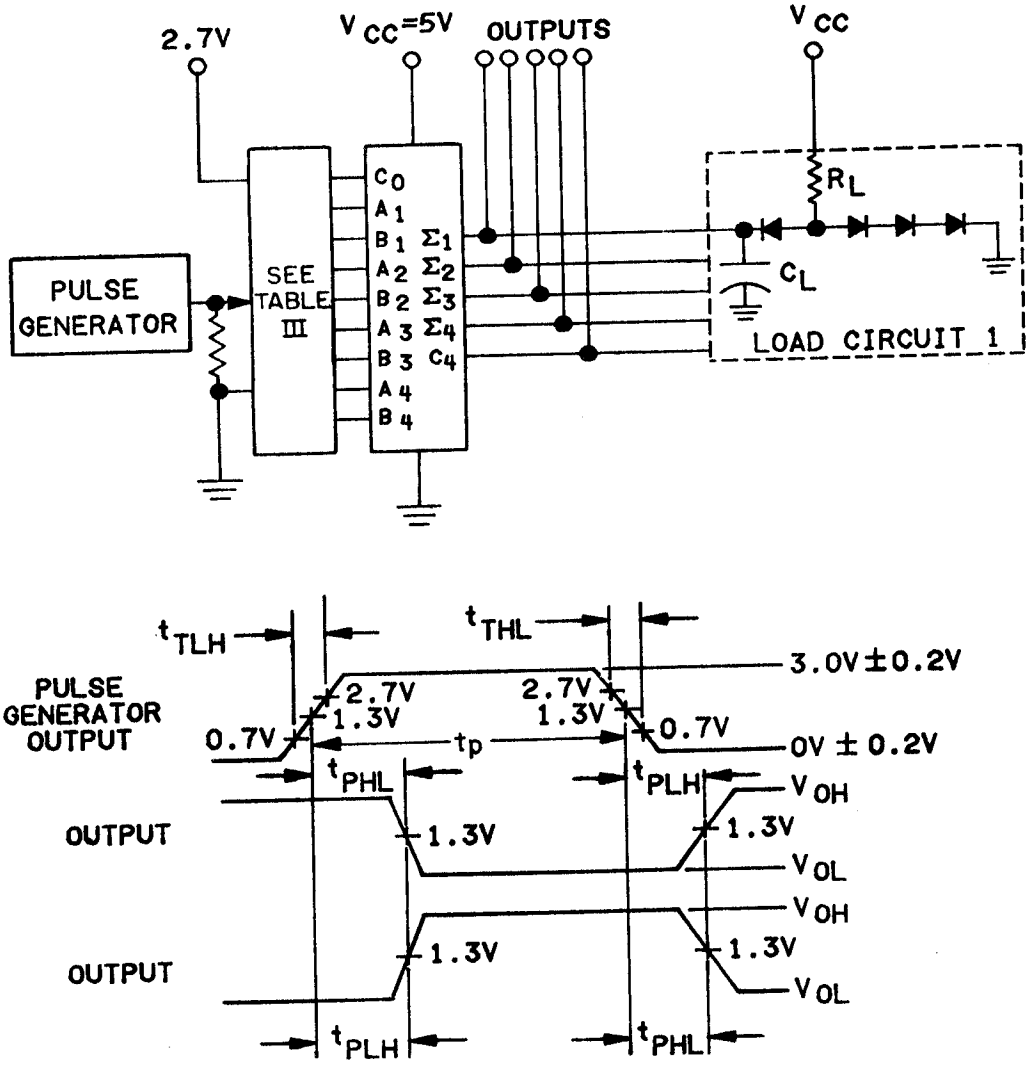


FIGURE 3. Logic diagrams - Continued.

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NOTES:
1. The pulse generator has the following characteristics: PRR $\leq$ 1 MHz, $t_{TLH} \leq$ 15 ns, $t_{THL} \leq$ 6 ns, t_p = 200 ns and $Z_{OUT} \cong$ 50Ω.
2. C_L = 50 pF ±10%, including scope probe, wiring, and stray capacitance without package in test fixture.
3. All diodes are 1N3064 or equivalent.
4. R_L = 2 kΩ ±5%.
5. Load circuits on a given output are only required when specified by "OUT" on the specific test in table III. Load circuits may otherwise be omitted.

FIGURE 5. Switching time test circuit for device types 01 and 02.

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Custodians:
Air Force - 17
Army - ER
Navy - EC

Review activities:
Air Force - 11, 19, 85, 99
Army - AR, MI
Navy - OS, SH, TD
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

(Project 5962-0799)

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